🛡️ ISEC2700 – Information Security

Mini-Project 1 (MP1): Small Business Security Risk Assessment

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1. Assignment Details

Field	Information
Assignment Title	Mini-Project 1 (MP1) – Small Business Security Risk Assessment
Course Code	ISEC2700
Course Title	Information Security
Instructor	Davis Boudreau
Assignment Type	Security Investigation / Risk Assessment
Weight	15%
Estimated Effort	4–6 hours
Delivery Mode	Individual Investigation
Due Date	See Brightspace
Submission Format	Professional Security Assessment Report (PDF)

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2. Overview / Purpose / Objectives

Overview

In this mini-project, you will conduct a **security investigation of a small business environment** and produce a professional **security risk assessment report**.

You will analyze a fictional organization, **MapleTech Accounting Inc.**, and identify weaknesses within its technology architecture.

Your task is to:

- identify security issues
- analyze risk
- prioritize vulnerabilities
- recommend security improvements

This assignment simulates the work performed by **cybersecurity analysts and security consultants** during an **architecture review and risk assessment engagement**.

Important Preparation

Before beginning this investigation, you **must carefully review the document:**

MP1 Foundations: Identifying Security Issues in a Small Business Environment

That document explains:

- how security professionals identify vulnerabilities
- how risk analysis works
- how risk assessment is performed
- how to analyze systems layer-by-layer

The investigation process introduced in the Foundations document will guide your work in this assignment.

You are expected to **reference the concepts from MP1 Foundations** throughout your report.

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3. Investigation Scenario

You are working as a **Junior Security Analyst** for a cybersecurity consulting firm.

Your team has been hired by **MapleTech Accounting Inc.** to perform a **security architecture review** and identify potential risks in their IT environment.

The organization wants to understand:

- where security weaknesses exist
- which risks are most serious
- which improvements should be prioritized

You will perform an **initial security assessment** using the information provided in this investigation packet.

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4. Organization Profile

Organization Name: MapleTech Accounting Inc. **Industry:** Accounting and Financial Services **Location:** Halifax Regional Municipality, Nova Scotia **Employees:** 22 **IT Support:** External part-time contractor

Business Overview

MapleTech Accounting Inc. provides financial services for small businesses and individuals.

Core services include:

- tax preparation

- bookkeeping
- payroll processing
- financial advisory services

Because MapleTech processes sensitive financial information, protecting data confidentiality and integrity is essential.

Sensitive Data Managed

Data Type	Sensitivity
Tax records	High
Payroll data	High
Client financial records	High
Internal accounting data	Medium
Employee HR records	Medium

Security failures could result in:

- financial fraud
- data breaches
- loss of client trust
- operational disruption

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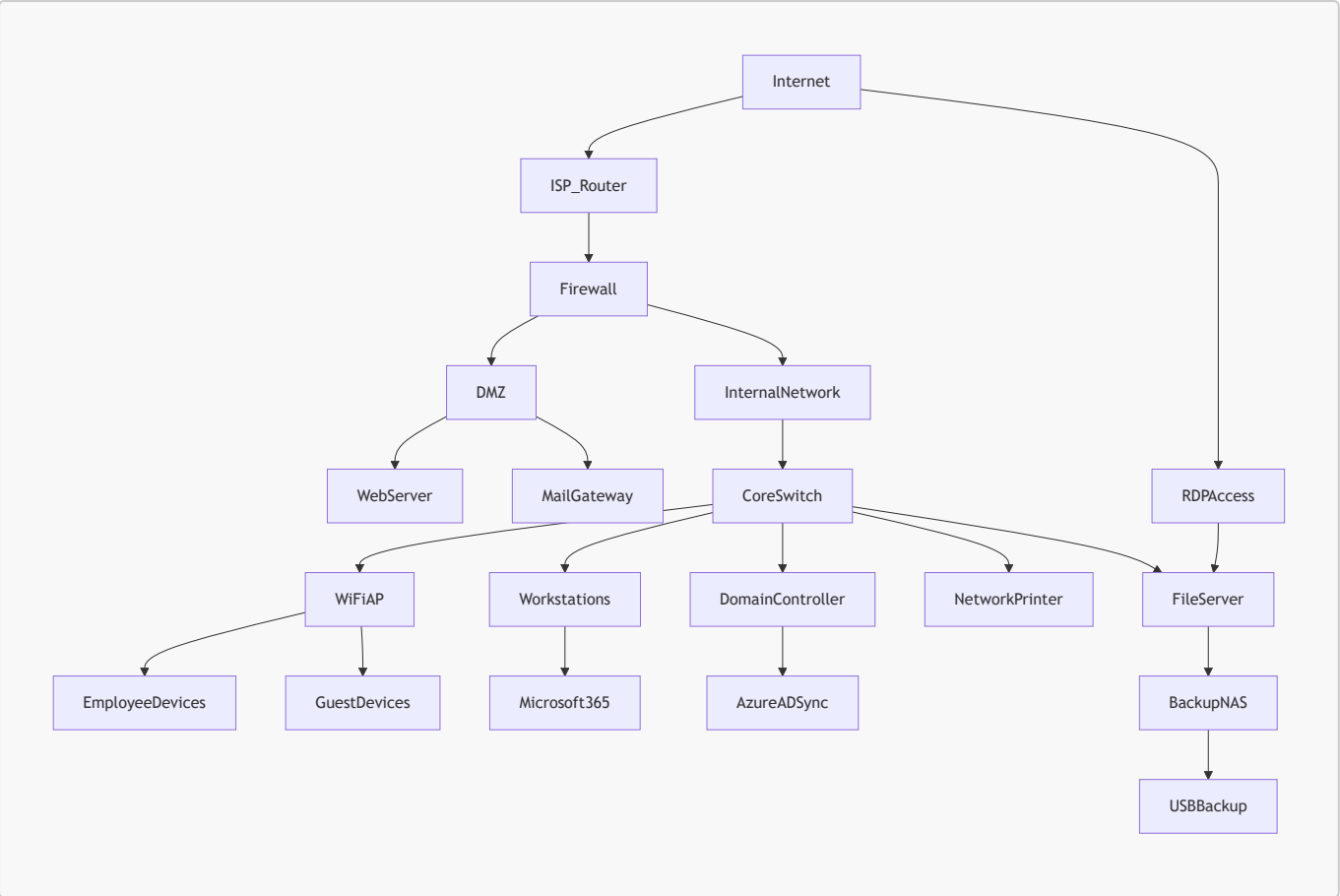
5. Technology Architecture Overview

MapleTech operates a small office network that combines **on-premise systems and cloud services**.

The architecture diagram below represents the current environment.



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6. Technology Inventory

The following infrastructure components exist within the MapleTech environment.

Network Infrastructure

System	Description
ISP Router	Internet connectivity
Firewall	Basic perimeter protection
Core Switch	Internal network connectivity
Wireless Access Point	Office wireless network

Servers

Server	Purpose
Domain Controller	Active Directory authentication
File Server	Shared document storage
Web Server	Public company website
Mail Gateway	Email filtering

End User Devices

Device Type	Quantity
Windows 11 Workstations	20
Laptops	4
Network Printer	1

Cloud Services

Service	Purpose
Microsoft 365	Email and collaboration
Azure AD Sync	Identity synchronization

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7. Operational Practices

User Accounts

Employees are issued:

- an Active Directory account
- a Microsoft 365 account

Authentication currently uses **username and password only**.

Password requirements:

- minimum length: 8 characters
- password expiration: every 180 days

Remote Access

Employees sometimes work remotely.

Remote connectivity is provided through:

- Remote Desktop (RDP)
- direct connection to internal systems

Wireless Access

The office has one wireless network used by:

- employees

- visitors
- contractors

The wireless password is occasionally shared with guests.

Backup Practices

The file server is backed up using the following process:

- daily backups to a **local NAS device**
- weekly backup copied to **USB storage**
- USB storage kept in the server room

Software Updates

Software updates are typically installed:

- when the IT contractor visits
- when users manually install updates

There is no automated patch management system.

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8. Security Controls Currently Implemented

The following controls are currently in place.

Control	Status
Firewall	Enabled
Antivirus	Installed on workstations
Windows updates	Installed periodically
Email spam filtering	Basic filtering

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9. Security Policies

MapleTech maintains a limited set of security guidelines.

Acceptable Use

Employees are expected to:

- use company computers for work purposes
- avoid suspicious websites

- report technical issues to the IT contractor

Password Policy

Employees must:

- use at least 8 characters
- change passwords every 180 days
- avoid sharing passwords

Data Storage Policy

Employees are instructed to store files on the **company file server** rather than local computers.

Incident Reporting

Employees should report suspicious emails or system problems to the office manager.

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10. Organizational Constraints

MapleTech has limited IT resources.

Current limitations include:

- no dedicated cybersecurity team
- limited IT budget
- one external IT contractor
- no centralized monitoring systems

The company wants to understand **which security improvements should be prioritized**.

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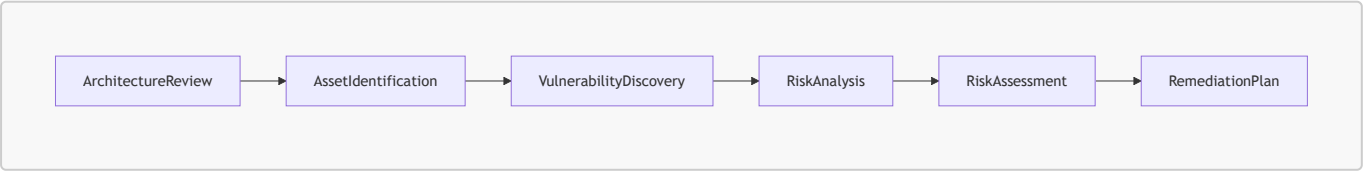
11. Your Assignment

Using the information in this packet, you must conduct a **security investigation**.

You should follow the **security analysis process introduced in the MP1 Foundations document**.



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12. Investigation Focus Areas

Your investigation should analyze the environment across multiple domains.

Domain	Focus
Perimeter Security	Internet-facing exposure
Network Architecture	Internal network design
Identity and Access	Authentication systems
Endpoint Security	Workstation protection
Data Protection	File storage and access
Backup and Recovery	Data resilience
Cloud Security	Microsoft 365 environment
Administrative Controls	Policies and procedures

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13. Final Deliverable

You must submit a **Professional Security Risk Assessment Report** that includes:

- Executive Summary
- Architecture Analysis
- Vulnerability Findings
- Risk Analysis
- Risk Matrix
- Top Risk Prioritization
- Remediation Plan

Your report should demonstrate that you understand how to **identify security issues and evaluate risk in a real organizational environment**.